

Collapse Pathways in Adaptive Temporal Abstraction (DreamerV3)

CIVO

IRD

Gradient Starvation

Trained dt -emb
 $dt > 1$ **imagination**

Trained dt -emb
 $dt = 1$ **imagination**

Frozen dt -emb
(no temporal signal)

Reward prediction bias
at $dt > 1$ ($2.78\times$ gap)

World model accuracy
ceiling at long horizons

Representation drift
(stale frozen features)

Value inflation ($NVE > 1.5\times$)
→ actor exploitation
entropy ↓ (over-committed)

value_std explosion
→ policy de-specialization
entropy ↑ (destabilizing)

actor_loss → 0
→ gradient starvation
NVE transient spike

CIVO Collapse
exp 008d, 006
NVE $> 1.5\times$ sustained

IRD Collapse
exp 008e_v3 s2
NVE passive rise

Starvation Collapse
exp 015b
NVE transient

C6 Correction: $dt = 1$ imagination

✓ Prevents CIVO mechanism

✗ Does NOT prevent late
policy instability (IRD)

[SGS] →
(proposed)